Forecasting and Macroeconomic Analytics

ECON 522

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Introduction

- History - popular in early 90 s, making comeback now.
- What is a Time Series? Y_t
  - The main difference between time series econometrics and cross-section is in dependence structure. Cross section econometrics mainly deals with i.i.d. observations, Y_i, while in time series each new arriving observation is stochastically depending on the previously observed, Y_t.
- The dependence is our best friend and a great enemy.
  - On one side, the dependence screw up your inferences (CLT you learned was for i.i.d. data). On the other side, the dependence allow us to do more by exploiting it. For example, we can make forecasts (which are almost non-sense in cross-section).

Introduction

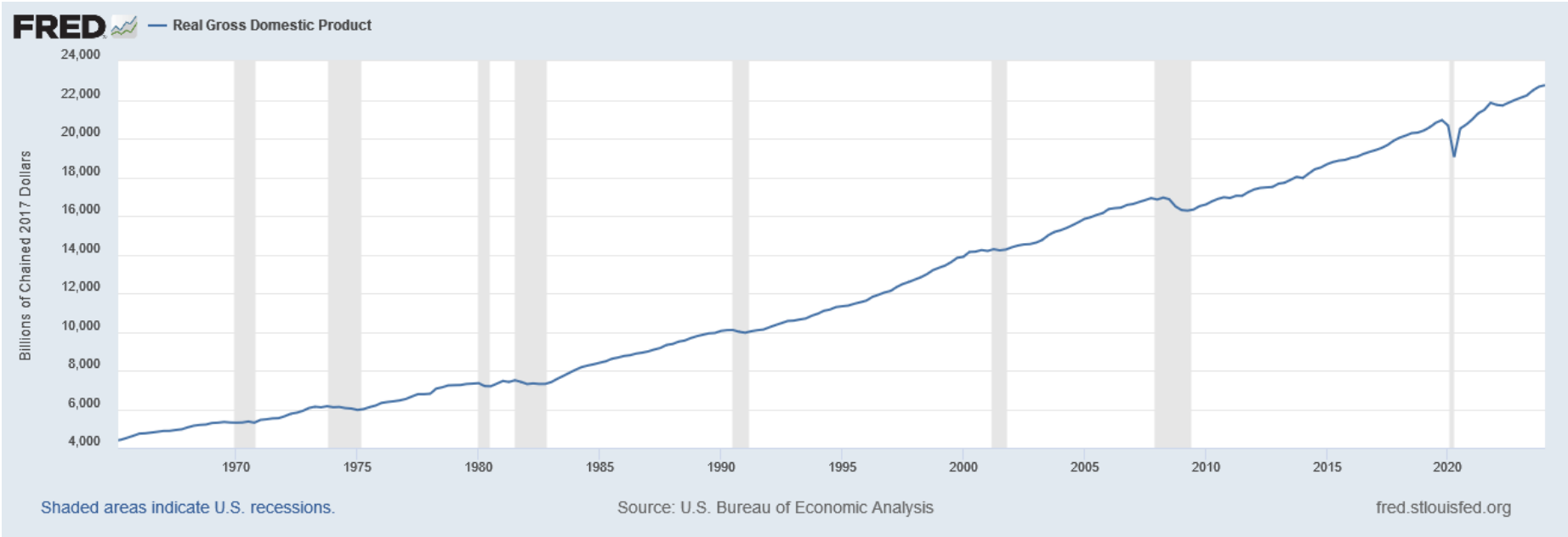
- We are going to ask questions like:
  - What are the characteristics of time series data?
  - Which type of questions can we ask/answer?
  - Which type of questions we cannot answer? Or can we?
- We can roughly divide time series into macro and finance related stuff.
  - Macro Time series mostly focuses on means. Often limited by small number of observations available over long horizon (e.g. 20 years monthly is T=300).
  - Financial data usually high-frequency over short period of time. This allows us to model volatility and higher moments.

Introduction

- Which type of questions can we ask/answer?
  - In the first part of the class we will be using Regression Models for *Forecasting*.
  - Forecasting and estimation of causal effects are quite different objectives (more on this later).
  - In the second half of the class we will discuss if and when we can talk about "causality".
- Forecasting:
  - It is very important in macro and finance.
  - Omitted variable bias isn't a problem.
  - External validity is paramount: the model estimated using historical data must hold into the (near) future
  - Intepretation of the coefficients is not the goal.

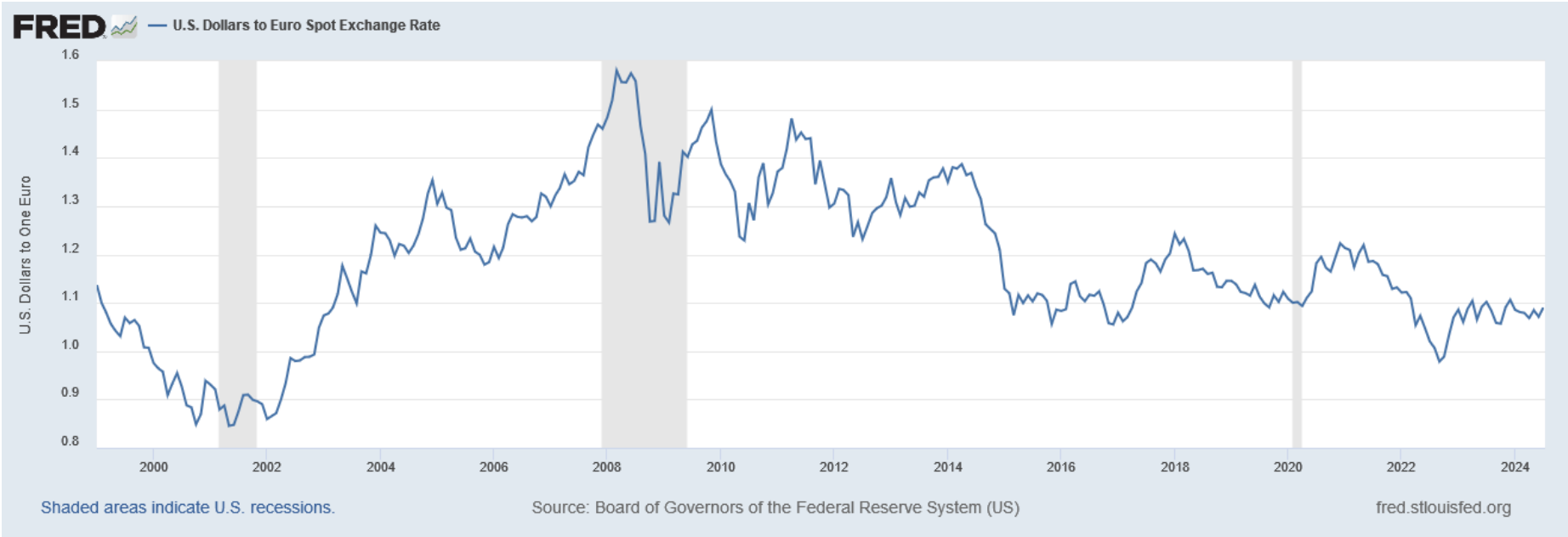
Some Examples of Time Series Data

Real GDP, 1965:1-2024:1, Quarterly (GDP is only quarterly), Seasonally adjusted, billions of chained 2017 dollars. From FRED.



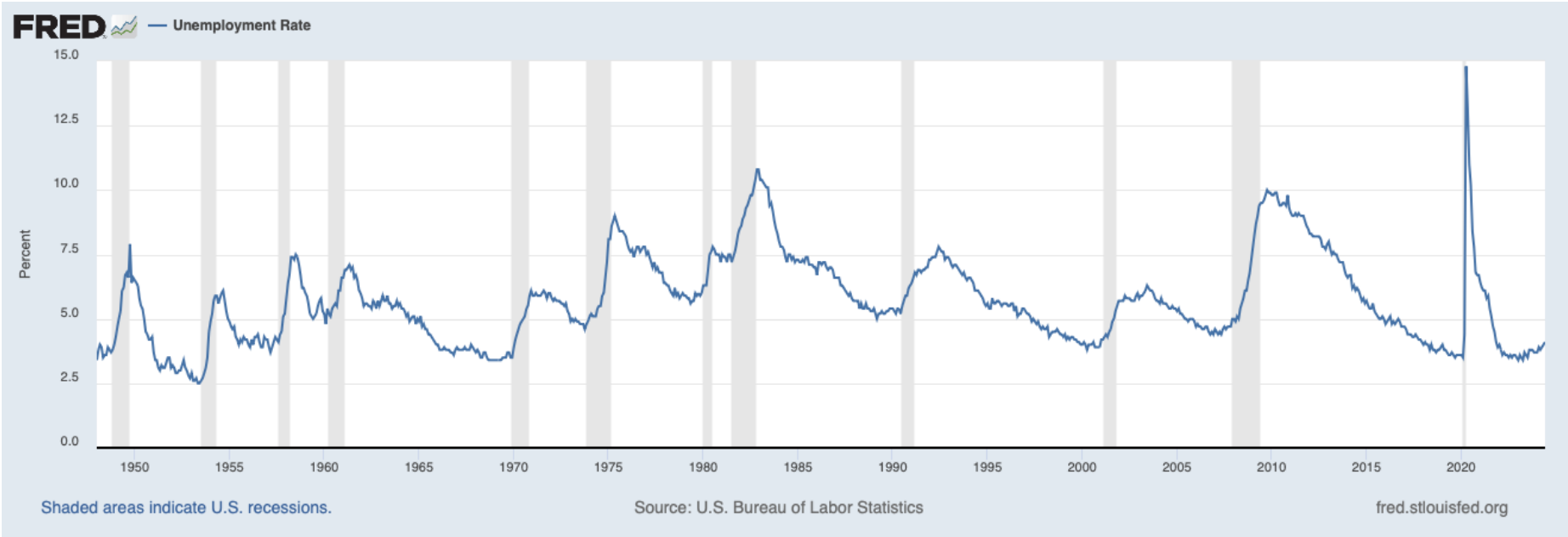
Some Examples of Time Series Data

US Dollars to Euro Spot Exchange rate. Not Seasonally Adjusted. 1999:01:04-2024:07:12. From FRED.



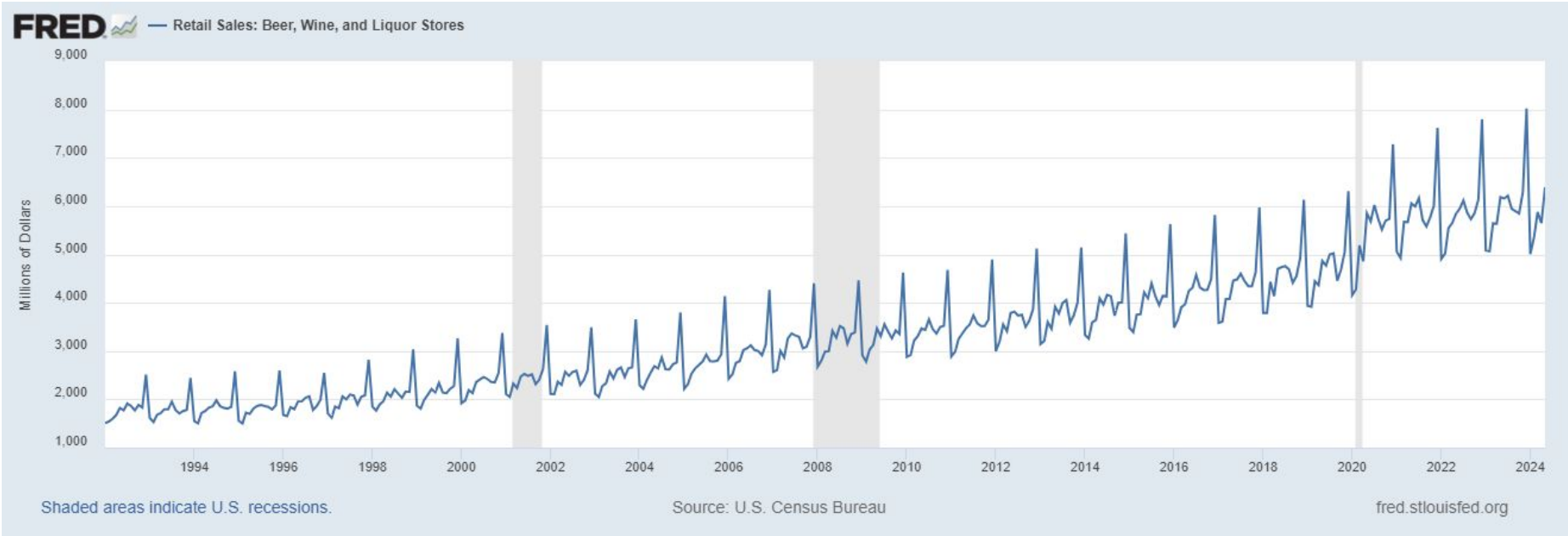
Some Examples of Time Series Data

Unemployment Rate (in %). Seasonally Adjusted. 1948:01-01 - 2024:06-01. From FRED.



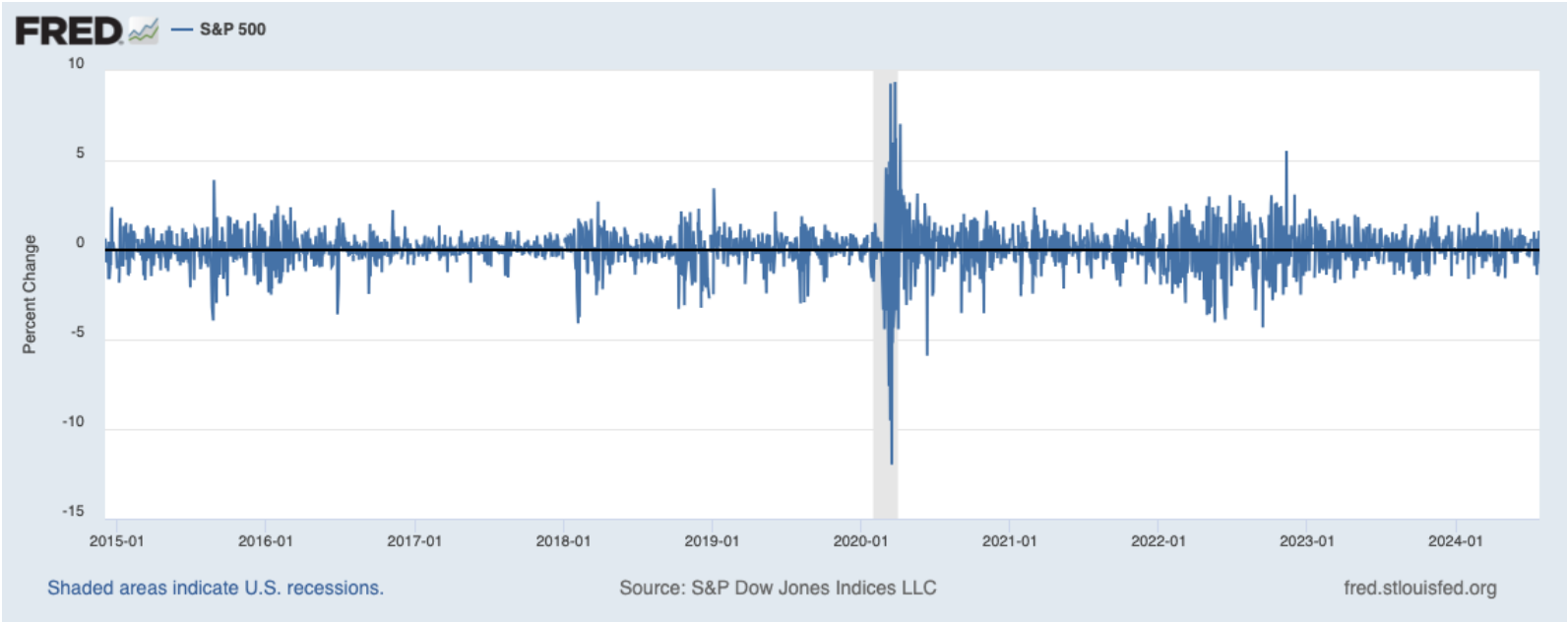
Some Examples of Time Series Data

- U.S. Liquor Retail Sales: Beer, Wine, and Liquor Stores
- Millions of Dollars, Not Seasonally Adjusted
- 1992-01-01 to 2024-05-01, Monthly



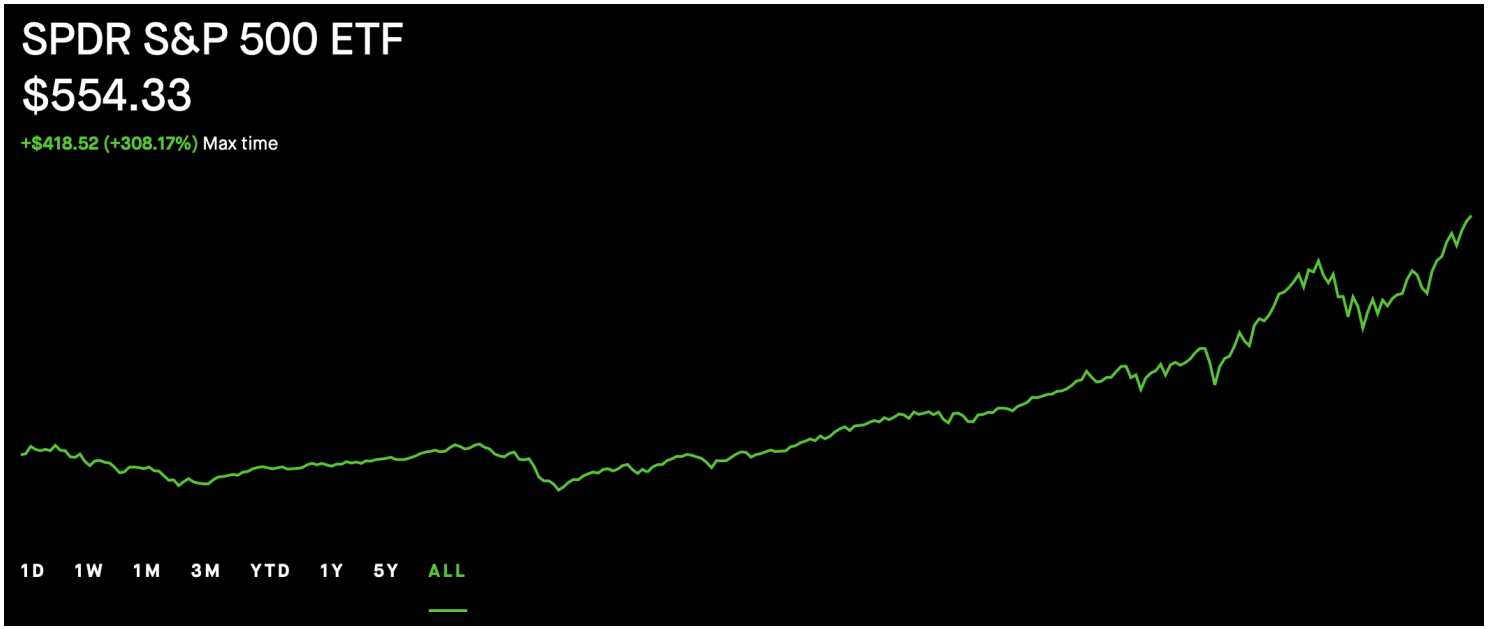
Some Examples of Time Series Data

- SP-500 returns
- Percentage Change, Not Seasonally Adjusted
- 2014-12-01 to 2024-07-23, Daily



Some Examples of Time Series Data

SPDR S&P 500 ETF (Ticker: SPY). Not Seasonally Adjusted.
2000:01-30 - 2024:07-21. From Robinhood



Time Series

Time series data raises new technical issues

- Time lags
- Correlation over time (serial correlation, a.k.a. autocorrelation)
- Forecasting models built on regression methods:
- autoregressive (AR) models
- autoregressive distributed lag (ADL) models
- Vector Autoregressive (VAR)
- need not (typically do not) have a causal interpretation
- Conditions under which dynamic effects can be estimated, and how to estimate them
- Calculation of standard errors when the errors are serially correlated

Components of any Time Series

There are **three** main components in any Time Series:

Trend

Part of a series' movement that corresponds to long-term, slow evolution

Seasonality

Part of a series' movement that repeats each year

Cycles

A catch-all phrase for various forms of dynamic behavior that link the present to the past and the future to the present

Road Map

1. Brief look at deterministic trend (it is easy) and seasonality.
2. How to model the cycles component will take a big part of this class. We will study this in univariate models first (ARMA models).
3. To be able to model and estimate cycles we need one important assumption (stationarity). We will need to define it and then see what happens when this assumption is not satisfied (non stationarity and unit root testing).
4. How to we use these models to forecast?
5. We will then see how to estimate cycles for multivariate models (VARMA)
6. We will then move to more advanced topics like bayesian estimation, stochastic variance etc.